

MODULE 02

STUDENT DOWNLOAD | SHOP-FLOOR WORKBOOK

How to Read Drawings Like a Fabricator

A drawing review checklist and red-flag worksheet for catching what the drawing does not openly show.

Engineers draw the finished condition. Fabricators pay for the path it takes to get there.

Use this immediately after watching the module. The goal is not paperwork for the sake of paperwork. The goal is to slow down the quote just enough to stop missed cost, unclear responsibility, and margin loss before the job is won.

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WORKSHEET

Drawing Read Sequence

Do not start with the biggest-looking view. Work through the set in a repeatable order.

STEP	WHAT TO CHECK	NOTES
1	Title block, drawing list, issue date, revision number	
2	General notes, structural notes, finish notes, spec references	
3	Plans, elevations, sections, details, schedules	
4	Material callouts: size, grade, thickness, finish, coatings, hardware	
5	Weld symbols, bolt notes, hole requirements, tolerances, inspection	
6	Field dimensions, site access, install sequence, crane/lift needs	
7	Conflicts, missing information, unclear responsibility, alternates	

WORKSHEET

Read the White Space

The drawing shows the finished condition. Your estimate must price the path to get there.

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HIDDEN ITEM	QUESTION TO ASK	NOTES
Handling	How does this piece get moved, flipped, staged, loaded, or unloaded?	
Fit-up	Will this need bracing, jigs, trial fit, special layout, or tight squaring?	
Welding / Finish	Are welds structural only, seal welds, appearance welds, ground smooth, or continuous?	
Coating	Is there blast, prime, paint, galvanizing, duplex coating, touch-up, or special prep?	
Shipping	Can it ship in one piece? Does it need to be broken down? Any oversized load issues?	
Install	Who provides anchors, layout, crane, lift, access, shutdowns, permits, and cleanup?	

WORKSHEET

Red-Light Details

These items may still be buildable, but they are not neutral. They usually change labour, risk, or schedule.

RED FLAG	IMPACT TO ESTIMATE	ACTION
Tight field fit or unknown dimensions	More layout, field verification, risk of rework	Clarify or include field measure
Special inspection / testing	More control time and documentation	Price and state requirement
Appearance grinding or smooth welds	More labour than structural welds	Price separately
Special material or certification	Availability, lead time, cost risk	Get supplier quote
Access or sequence restrictions	Install labour and equipment risk	Qualify in quote
Repeated inefficient detail	High savings potential	Offer alternate

WORKSHEET

Alternate Opportunity

A strong estimator does not just complain about bad details. They look for buildable options.

AS DRAWN ISSUE	RECOMMENDED ALTERNATE	POSSIBLE SAVINGS / BENEFIT

WORKSHEET

Apply This To A Real Job

Use the space below to connect the lesson to an actual estimate you are working on.

APPLY THIS TO MY NEXT ESTIMATE	NOTES
One thing I will use immediately	
One question I need to ask before I price	
One assumption, exclusion, or clarification I should write down	

APPLY THIS TO MY NEXT ESTIMATE	NOTES
Extra notes / reminders	